

Lab04_LuizCelin

Luiz Fernando Celin

Questão 1

```
library(readxl)
library(tidyverse)
```

Warning: pacote 'tidyverse' foi compilado no R versão 4.5.3

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Questão 2

```
fname <- 'WDIEXCEL.xlsx'
file.exists(fname)
```

```
[1] TRUE
```

Questão 3

```
countries <- read_excel(fname, sheet = "Country")
```

Questão 4

```
countries <- countries |>  
  filter(`Country Code` %in% c("BRA", "USA", "CHN", "AGO", "COL"))
```

Questão 5

```
WDI <- read_excel(fname, sheet = "Data")
```

Questão 6

```
WDI <- WDI |>  
  select(`Country Code`, `Indicator Code`, c(`1960`:`2018`))
```

Questão 7

```
WDI <- WDI |>  
  pivot_longer(  
    cols = `1960`:`2018`,  
    names_to = "YEAR",  
    values_to = "Values"  
  )
```

Questão 8

```
is.integer(WDI$YEAR)
```

[1] FALSE

```
WDI <- WDI |>  
  mutate(YEAR = as.integer(YEAR))  
  
is.integer(WDI$YEAR)
```

[1] TRUE

Questão 9

```
WDI <- WDI |>  
  inner_join(  
    countries |> select(`Country Code`, `Short Name`),  
    by = "Country Code") |>  
  select(`Country Code`, `Short Name`, `YEAR`, `Indicator Code`, `Values`) |>  
  filter(`Indicator Code` == "SP.DYN.CBRT.IN")
```

Questão 10

```
WDI <- WDI |>  
  arrange(`Country Code`, YEAR)
```

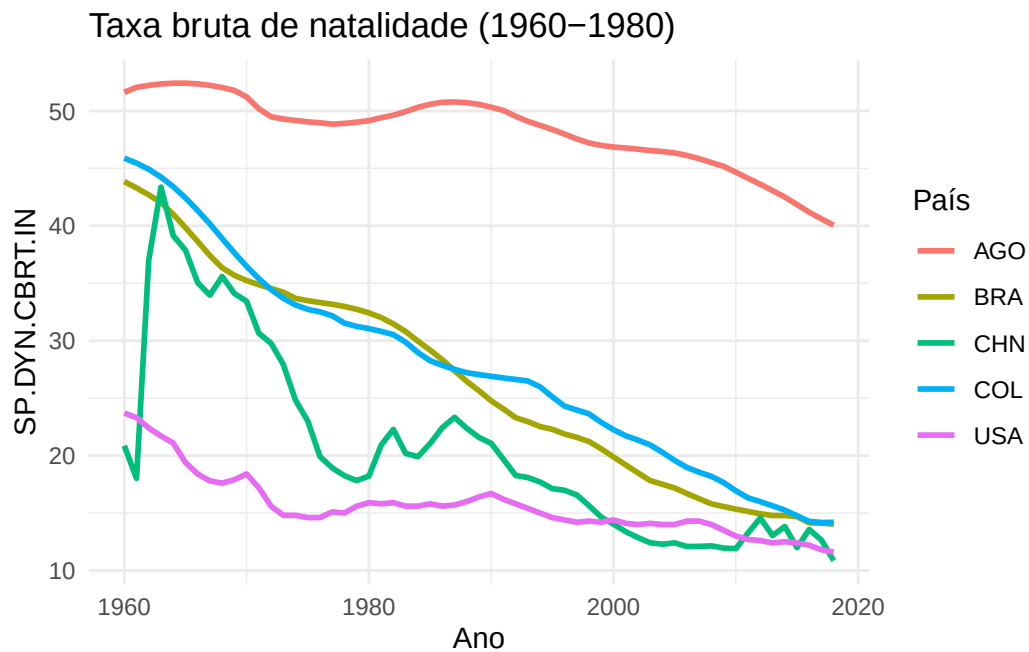
questão 11

```
library(ggplot2)  
  
ggplot(WDI, aes(x = YEAR, y = `Values`, color = `Country Code`)) +  
  geom_line(linewidth = 1) +  
  labs(
```

```

title = "Taxa bruta de natalidade (1960-1980)",
x = "Ano",
y = "SP.DYN.CBRT.IN",
color = "País"
) +
theme_minimal()

```



```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-09 23:36:40 -03"
```